

Rishu

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 GitHub

 LinkedIn

 Kaggle

OVERVIEW

AI Engineer and Backend Developer with professional experience at TCS and hands-on expertise in building Agentic AI systems, RAG applications, microservices architectures, and scalable backend platforms using FastAPI, Docker, Ollama, SQL, and REST APIs. Skilled in Generative AI, LLM integration, production-ready backend development.

EXPERIENCE

• Tata Consultancy Services (TCS)

Operations & Technology Analyst

Aug 2025 – Present

- Worked in the Tax and Invoice Processing team handling global invoice operations.
- Extracted and processed financial and invoice data from enterprise systems using Oracle Applications and Sabrix.
- Performed data validation, tax classification, and compliance checks across invoices from multiple international regions.
- Automated invoice processing workflows using Generative AI models, reducing manual processing effort by 65%.
- Collaborated with cross-functional teams to streamline financial operations and improve processing accuracy.

PROJECTS

• Microservices Powered Backend Ecommerce

Github 

- Developed distributed microservices using FastAPI for inventory, payment, and authentication services.
- Implemented API Gateway with JWT-based authentication and secure service routing.
- Used Docker Compose for container orchestration and Redis/MySQL for backend storage.
- Designed scalable REST APIs following modular backend architecture principles.
- Worked with automated API testing, debugging, and service integration across multi-tier systems.

• ATS Resume Scorer using LLMs

- Built an ATS evaluation platform that extracts resume information from PDF documents and generates structured candidate profiles.
- Utilized Large Language Models to perform resume parsing, skill extraction, and ATS score generation.
- Developed REST APIs using FastAPI for resume upload, scoring, and profile generation.
- Implemented JSON-based candidate evaluation workflows for recruiter-friendly candidate assessment.
- Designed modular architecture supporting multiple LLM backends and scalable deployment.

• RAG Agentic Chat System using FastAPI and Ollama

- Built an Agentic AI-powered RAG application using FastAPI and locally hosted Ollama (LLaMA 3.2) for intelligent document retrieval and conversational querying.
- Designed RESTful APIs for semantic search, document indexing, and context-aware response generation.
- Implemented retrieval pipelines and tool-based agent workflows inspired by LangChain concepts for enterprise document interaction.
- Integrated MySQL for persistent chat/session storage and automated ingestion pipelines.
- Containerized services using Docker for scalable deployment and modular backend architecture.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL

AI/ML: Supervised Learning, Unsupervised Learning, NLP, Feature Engineering, Model Evaluation, Hyperparameter Tuning

Generative AI: RAG Pipelines, Agentic AI, Prompt Engineering, LLM Applications, Semantic Search

Frameworks/Tools: FastAPI, Docker, LangChain, LangGraph, Linux, Git/GitHub, REST APIs, Ollama

Libraries: Scikit-learn, NLTK, Pandas, NumPy

Databases: MySQL, PostgreSQL, Redis

Software Engineering: OOP, Microservices, API Development, JWT Authentication, API Gateway

Testing & Deployment: Dockerized Deployment, API Testing, Backend Debugging

Areas of Interest: Generative AI, Agentic Systems, NLP, Applied Machine Learning

EDUCATION

- **Heritage Institute of Technology, Kolkata** 2021 – 2025
B.Tech in CSE (AI & ML) | CGPA: 8.06
- **Saraswati Shishu Vidya Mandir** 2020
Board of Higher Secondary Education | Percentage: 93.4%
- **Saraswati Shishu Vidya Mandir** 2018
Board of Secondary Education | Percentage: 84%

ACHIEVEMENTS

- **Kaggle ML Olympiad** *Secured 27th rank by developing a predictive model for earthquake damage assessment.*
- **House Price Prediction Challenge** *Achieved top 10 percentile by building an accurate regression model.*
- **Coding Competition Organizer** *Organized a hostel coding competition with 100+ participants focused on algorithms and problem-solving.*